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### Information processing apparatus, moving object, system, information processing method, and computer-readable storage medium

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#### Abstract

There is provided an information processing apparatus including: a receiving control unit configured to perform a control to receive information indicating a risk area ahead in a movement direction of a moving object, from a server configured to retain information relating to a risk area; a prediction unit configured to predict a change in the movement direction of the moving object; and a determination unit configured to determine whether the movement direction of the moving object has been changed to the direction predicted by the prediction unit, in which the receiving control unit is configured to perform a control to receive the information indicating the risk area, when the determination unit determines that the movement direction of the moving object has been changed to the direction predicted by the prediction unit.

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References Cited

U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl. | CPC        |
|--------------|-------------|---------------|----------|------------|
| 2019/0088115 | 12/2018     | Tanabe        | N/A      | N/A        |
| 2021/0281968 | 12/2020     | Kurehashi     | N/A      | N/A        |
| 2022/0194368 | 12/2021     | Nemoto        | N/A      | H04W 4/46  |
| 2023/0298468 | 12/2022     | Jha           | 701/117  | G08G 1/164 |

FOREIGN PATENT DOCUMENTS

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 109155108  | 12/2018          | CN      | N/A |
| 2018181088 | 12/2017          | JP      | N/A |
| 2021140470 | 12/2020          | JP      | N/A |
| 2019003602 | 12/2018          | WO      | N/A |
| 2021005392 | 12/2020          | WO      | N/A |

OTHER PUBLICATIONS

“A scheduling algorithm for autonomous driving tasks on mobile edge computing servers,” Hongjun Dai, Xiangyu Zeng, Zhilou Yu, Tingting Wang, Journal of Systems Architecture, p. 14-23, whole document (Year: 2019). cited by examiner

“Performance of Vehicle-to-Vehicle Communication using IEEE 802.11p in Vehicular Ad-hoc Network Environment,” Vaishali D. Khairnar, Dr. Ketan Kotecha, Research Scholar Institute of Technology Nirma University Ahmadabad, p. 143-170, whole document. (Year: 2013). cited by examiner

Office Action issued for counterpart Chinese Application 202211657272.X, issued by The State Intellectual Property Office of People's Republic of China on Jul. 29, 2025. cited by applicant

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Background/Summary

(1) The contents of the following Japanese patent application(s) are incorporated herein by reference: NO. 2022-008744 filed in JP on Jan. 24, 2022.

## BACKGROUND

### 1. Technical Field

(2) The present invention relates to an information processing apparatus, a moving object, a system, an information processing method, and a computer-readable storage medium.

### 2. Related Art

(3) Patent document 1 discloses a system in which an MEC server manages a risk area and the MEC server provides information of the risk area to each vehicle.

### Prior Art Document

(4) Patent Document 1: Japanese Patent Application Publication No. 2021-140470

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 schematically shows a scene where a system **10** is used.

(2) FIG. 2 shows a situation in which an information processing apparatus **64** included in a vehicle **60** requests location information of a risk area.

(3) FIG. 3 shows a situation in which the information processing apparatus **64** makes an inquiry relating to the risk area.

(4) FIG. 4 shows a system configuration of a vehicle **20**.

(5) FIG. 5 shows a system configuration of a server **52**.

(6) FIG. 6 shows an outline of an operation when the vehicle **20** transmits a risk area request to the server **52** according to a change in a travel path.

(7) FIG. 7 shows an example of a flowchart relating to processing for an information processing apparatus **24** to receive coordinate information of the risk area.

(8) FIG. 8 shows an example of a computer **2000**.

### DESCRIPTION OF EXEMPLARY EMBODIMENTS

(9) Hereinafter, (some) embodiment(s) of the present invention will be described. The embodiment(s) do(es) not limit the invention according to the claims. In addition, not all of the combinations of features described in the embodiments are essential to the solution of the invention.

(10) FIG. 1 schematically shows a scene where a system **10** is used. The system **10** includes a vehicle **20**, a vehicle **60**, a terminal **82**, a base station **50**, and a server **52**.

(11) The vehicle **20** and the vehicle **60** are vehicles traveling on a road **70**. The vehicle **20** and the vehicle **60** are examples of a moving object. The vehicle **20** includes an information processing apparatus **24** and a sensor **29**. The sensor **29** is configured to include a camera. The information processing apparatus **24** has a function of processing information acquired by the sensor **29**, and a communication function with the server **52**. The vehicle **20** is, for example, a vehicle that has an advanced driver assistance system (ADAS) function. The vehicle **60** includes an information processing apparatus **64**. The information processing apparatus **64** has the communication function with the server **52**. The vehicle **60** is, for example, a vehicle that does not have the ADAS function.

(12) The terminal **82** is a terminal possessed by a person **80**. The terminal **82** is, for example, a mobile terminal such as a smartphone. The base station **50** is a base station for a mobile communication. The server **52** is a server connected to the base station **50**. The server **52** may include, for example, an edge computing server such as a mobile edge computing (MEC) server. The server **52** continuously manages location information of the terminal **82**. It should be noted that one server **52** is illustrated in FIG. 1, but the server **52** may be constituted by a plurality of servers respectively connected to a plurality of base stations on a one to one basis. The information

processing apparatus **24** may communicate with a server near the vehicle **20** among the plurality of servers constituting the server **52**, and the information processing apparatus **64** may communicate with a server near the vehicle **60** among the plurality of servers constituting the server **52**.

(13) In FIG. **1**, the vehicle **20** and the vehicle **60** are vehicles travelling along the road **70**. A vehicle **90** is a vehicle parked on the road **70**. The vehicle **60** is traveling behind the vehicle **20** in the same advance direction as the vehicle **20**.

(14) For the vehicle **20**, an area **110** on an advance direction side of the vehicle **20** further than the parked vehicle **90** is an area that is difficult to visually recognize from a location of the vehicle **20**. The information processing apparatus **24** identifies, as a risk area, the area **110** that is out of visibility from the vehicle **20**, from information such as an image which is ahead in an advance direction and is acquired by the sensor **29**.

(15) For example, the information processing apparatus **24** determines total four vertices of a vertex **111**, a vertex **112**, a vertex **113**, and a vertex **114** of the rectangular area **110** including a location of the vehicle **90** based on recognition information of the image acquired by the sensor **29**. The vertex **113** is a point away from the vertex **111** by a distance **L1** in the advance direction of the vehicle **20**, the vertex **111** being determined based on the recognition information of the image. The vertex **114** is a point away from the vertex **112** by the distance **L1** in the advance direction of the vehicle **20**, the vertex **112** being determined based on the recognition information of the image. The **L1** is a distance determined according to a vehicle speed of the vehicle **20**. The information processing apparatus **24** determines the vertex **111** and the vertex **113** such that a line segment connecting the vertex **111** and the vertex **113** is parallel to the advance direction of the vehicle **20** and that the line segment connecting the vertex **111** and the vertex **113** is located outside a region where the vehicle **90** recognized by the sensor **29** is present. In addition, the information processing apparatus **24** determines the vertex **111** and the vertex **112** such that a line segment connecting the vertex **111** and the vertex **112** is substantially orthogonal to the advance direction of the vehicle **20** and that the line segment connecting the vertex **111** and the vertex **112** is located outside the region where the vehicle **90** recognized by the sensor **29** is present.

(16) When the information processing apparatus **24** inquires of the server **52** whether the terminal **82** is present within the area **110**, the information processing apparatus **24** transmits, to the server **52**, risk area information including the total four vertices of the vertex **111**, the vertex **112**, the vertex **113**, and the vertex **114**. In the example of FIG. **1**, no terminal is present in the area **110** defined by the total four vertices of the vertex **111**, the vertex **112**, the vertex **113**, and the vertex **114**, and thus the server **52** either discards the risk area information, or transmits, to the vehicle **20**, response information to the effect that the terminal **82** is not present. The server **52** stores the total four vertices of the vertex **111**, the vertex **112**, the vertex **113**, and the vertex **114** which are included in the risk area information received from the information processing apparatus **24**.

(17) FIG. **2** shows a situation in which an information processing apparatus **64** included in a vehicle **60** requests location information of a risk area. When the information processing apparatus **64** is present in an area where a communication with the server **52** is possible, the information processing apparatus **64** transmits a risk area request to the server **52**. The server **52** transmits, to the information processing apparatus **64**, stored coordinate information of the vertex **111** and the vertex **112** of the area **110**, as response information to the risk area request. The information processing apparatus **64** stores the coordinate information of the vertex **111** and the vertex **112** received from the server **52**.

(18) FIG. **3** shows a situation in which the information processing apparatus **64** makes an inquiry relating to the risk area. When a distance of the vehicle **20** to a location which is indicated by the coordinate information of at least one of the vertex **111** or the vertex **112**, is less than a predetermined distance, the information processing apparatus **64** set an area **120** that is defined by the vertex **111**, the vertex **112**, a vertex **123**, and a vertex **124**, as a risk area for the vehicle **60**. For example, the information processing apparatus **64** sets, as the vertex **123**, a point away from the

vertex **111** in an advance direction of the vehicle **60**, by a distance **L2** determined according to a vehicle speed of the vehicle **60**, and sets, as the vertex **124**, a point away from the vertex **112** in the advance direction of the vehicle **60**, by the distance **L2**. The information processing apparatus **64** determines the vertex **123** and the vertex **124**, in a method similar to a method in which the information processing apparatus **24** determines the vertex **113** and the vertex **114**.

(19) The information processing apparatus **64** transmits, to the server **52**, the risk area information including coordinate information of the vertex **111**, the vertex **112**, the vertex **123**, and the vertex **124** of the area **120** set as the risk area. The server **52** transmits warning information to the information processing apparatus **64** and the terminal **82**, when the location information of the terminal **82** that is managed by the server **52** is included in the area **120** surrounded by the vertices which are included in the risk area information. When the warning information is received from the server **52**, the information processing apparatus **64** outputs a warning to an occupant of the vehicle **60**. For example, the information processing apparatus **64** outputs the warning to the occupant through an HMI (Human Machine Interface) function of the vehicle **20**. This makes it possible for the information processing apparatus **64** to output the warning by using the risk area information received from the server **52** by a wireless communication function, even though the information processing apparatus **64** itself has a function of recognizing the risk area by sensing means such as a camera. In addition, when the warning information is received from the server **52**, the terminal **82** outputs the warning to the person **80**. For example, the terminal **82** outputs the warning to the person **80** through the HMI function of the terminal **82**.

(20) In this way, the server **52** can collect the coordinate information of the risk area from the vehicle, and provide the coordinate information of the risk area to another vehicle. In FIG. **4** and subsequent figures, a description will be made mainly by using processing of the vehicle **20** acquiring the coordinate information of the risk area from the server **52**.

(21) FIG. **4** shows a system configuration of a vehicle **20**. The vehicle **20** includes the sensor **29**, the information processing apparatus **24**, a communication apparatus **48**, an information output apparatus **40**, and a turn signal lamp **42**.

(22) The sensor **29** includes a radar **21**, a camera **22**, a GNSS receiving unit **25**, a vehicle speed sensor **26**, and a yaw rate sensor **28**. The radar **21** may be a LiDAR, a millimeter wave radar, or the like. The GNSS receiving unit **25** receives a radio wave transmitted from a GNSS (Global Navigation Satellite System) satellite. The GNSS receiving unit **25** generates information indicating a current location of the vehicle **20** based on a signal received from the GNSS satellite. The camera **22** is an example of an image capturing apparatus that is mounted on the vehicle **20**. The camera **22** captures an image around the vehicle **20** to generate image information. For example, the camera **22** captures the image ahead in the advance direction of the vehicle **20** to generate the image information. The camera **22** may be a monocular camera. The camera **22** may be a compound eye camera, and may be a camera capable of acquiring distance information to an object. The camera **22** recognizes the object based on an image acquired by an image capturing function, and outputs location information of the recognized object. The vehicle speed sensor **26** detects the vehicle speed of the vehicle **20**. The yaw rate sensor **28** detects a rate of change of a rotation angle of the vehicle **20** in a turning direction.

(23) It should be noted that the vehicle **20** may include a driver assistance control apparatus that uses the information detected by the sensor **29** to perform driver assistance for the vehicle **20**. The driver assistance control apparatus may be realized by an ECU that provides the ADAS function.

(24) The communication apparatus **48** is responsible for the communication with server **52**. The communication apparatus **48** may communicate with the server **52** by the mobile communication. The communication apparatus **48** may be capable of communicating, for example, through a communication (Uu) interface via a portable base station for a vehicle-to-vehicle communication.

(25) The information output apparatus **40** is an apparatus that outputs alert information. The information output apparatus **40** may have the HMI function. The information output apparatus **40**

may include a head-up display or a navigation system. The information output apparatus **40** may be a mobile terminal possessed by the occupant of the vehicle **20**. The information output apparatus **40** may include a voice output apparatus that outputs the alert information by a voice. The turn signal lamp **42** blinks according to an operation of the occupant of the vehicle **20** or the like.

(26) The information processing apparatus **24** includes a control unit **200** and a storage unit **280**. The control unit **200** is realized, for example, by a circuit of a calculation processing apparatus or the like including a processor. The storage unit **280** is realized by including a non-volatile storage medium. The control unit **200** performs processing by using information stored in the storage unit **280**. The control unit **200** may be realized by an ECU (Electronic Control Unit) including a microcomputer including a CPU, a ROM, a RAM, an I/O, a bus, and the like.

(27) The control unit **200** includes a coordinate information acquisition unit **210**, a risk area identification unit **220**, a prediction unit **230**, a determination unit **240**, a control unit **208**, a transmission control unit **250**, and a receiving control unit **260**. It should be noted that a mode in which the control unit **200** does not have some function of functional blocks shown in FIG. 4, may be adopted. For example, a mode in which only some function is implemented in the control unit **200**, and another function is implemented as a function of the sensor **29** or another circuit, or the like, may be adopted.

(28) The risk area identification unit **220** is configured to identify the risk area outside the vehicle **20**. The transmission control unit **250** is configured to perform a control to transmit the risk area information indicating the risk area identified by the risk area identification unit **220**, to the server **52** configured to retain information relating to the risk area.

(29) It should be noted that the risk area may be an area that is outside the vehicle **20** and that poses a risk according to a movement of the vehicle **20**. The risk area may be an area that is out of visibility from the location of the vehicle **20** by an object outside the vehicle **20**. Information of the area that is out of visibility is, for example, location information of an area which is blocked for an occlusion to occur by a three-dimensional object such as another vehicle, a building, and a roadside tree, when viewed from the location of the vehicle **20**.

(30) The coordinate information acquisition unit **210** acquires the coordinate information of the object recognized from the image of the outside of the vehicle **20**, the image being captured by the camera **22** mounted on the vehicle **20**. The risk area identification unit **220** may identify the risk area based on the coordinate information of the object recognized from the image of the outside of the vehicle **20**, the image being captured by the camera **22**.

(31) The receiving control unit **260** is configured to perform a control to receive information indicating the risk area ahead in a movement direction of the vehicle **20**, from the server **52** configured to retain the information relating to the risk area. The prediction unit **230** is configured to predict a change in the movement direction of the vehicle **20**. The prediction unit **230** is configured to predict the change in the movement direction of the vehicle **20** based on operation information of the turn signal lamp **42** included in the vehicle **20**. The determination unit **240** is configured to determine whether the movement direction of the vehicle **20** has been changed to the direction predicted by the prediction unit **230**. The receiving control unit **260** is configured to perform a control to receive the information indicating the risk area, when the determination unit **240** determines that the movement direction of the vehicle **20** has been changed to the direction predicted by the prediction unit **230**.

(32) The receiving control unit **260** is configured to perform a control to receive the information indicating the risk area at a predetermined time interval. The receiving control unit **260** is configured to perform a control to receive the information indicating the risk area, at a timing other than a timing at which the information indicating the risk area is received at the predetermined time interval, when the determination unit **240** determines that the movement direction of the vehicle **20** has been changed to the direction predicted by the prediction unit **230**.

(33) The receiving control unit **260** is configured to perform a control to discard already received

information indicating the risk area, and receive the information indicating the risk area, when the determination unit **240** determines that the movement direction of the vehicle **20** has been changed to the direction predicted by the prediction unit **230**.

(34) The determination unit **240** is configured to determine that the movement direction of the vehicle **20** has been changed to the direction predicted by the prediction unit **230**, in a case where an operation of the turn signal lamp **42** is stopped and then the vehicle **20** travels a distance greater than or equal to a predetermined distance. The determination unit **240** is configured to store the movement direction of the vehicle **20** in a case where the operation of the turn signal lamp **42** is started, and the determination unit **240** is configured to determine that the movement direction of the vehicle **20** has been changed to the direction predicted by the prediction unit **230**, in a case where the operation of the turn signal lamp **42** is stopped and then the vehicle **20** travels a predetermined distance or more in a direction different from the stored movement direction.

(35) The determination unit **240** may store the movement direction of the vehicle **20** in a case where the operation of the turn signal lamp **42** is started, and the determination unit **240** may determine that the movement direction of the vehicle **20** has been changed to the direction predicted by the prediction unit **230**, in a case where the operation of the turn signal lamp **42** is stopped and then the vehicle **20** travels a predetermined time or more in a direction different from the stored movement direction.

(36) It should be noted that the “movement direction of the vehicle **20**” may be a travel direction of the vehicle **20**. The acquisition and the prediction of the travel direction of the vehicle **20** by the prediction unit **230** and the determination unit **240** may be performed based on any combination of the GNSS signal received by the GNSS receiving unit, a signal which is output by the yaw rate sensor **28**, and the operation information of the turn signal lamp **42**.

(37) The control unit **208** may control the driver assistance for the vehicle **20** or an execution of the warning to the occupant of the vehicle **20**. For example, when the information output apparatus **40** includes a head-up display, the control unit **208** may cause the head-up display of the vehicle **20** to output light for forming a mark as alert information indicating that a pedestrian is present in the risk area. In addition, the control unit **208** causes the head-up display to output the light for forming the mark in a display region corresponding to a location of the risk area where the pedestrian is present. The control unit **208** may project the light for forming the mark toward a reflective member provided on a windshield of the vehicle **20**. It should be noted that the control unit **208** may output the alert information by a voice or a character. In addition, the control unit **208** may control the travel of the vehicle **20** through the driver assistance control apparatus included in the vehicle **20**.

(38) FIG. 5 shows a system configuration of a server **52**. The server **52** includes a communication apparatus **348**, a control unit **300**, and a storage unit **380**.

(39) The control unit **300** controls the communication apparatus **348**. The communication apparatus **348** is responsible for a communication between the terminal **82** and the information processing apparatus **24**. The control unit **300** is realized, for example, by a circuit of a calculation processing apparatus or the like including a processor. The storage unit **380** is realized by including a non-volatile storage medium. The control unit **300** performs processing by using information stored in the storage unit **380**. The control unit **300** may be realized by a microcomputer including a CPU, a ROM, a RAM, an I/O, a bus, and the like.

(40) The control unit **300** includes a retention control unit **310**, a determination unit **320**, a transmission control unit **350**, and a receiving control unit **360**. It should be noted that a mode in which the control unit **300** does not have some function of functional blocks shown in FIG. 5, may be adopted.

(41) The receiving control unit **360** performs a control to receive the risk area information indicating the risk area outside the vehicle **20** identified by the vehicle **20**. The retention control unit **310** performs a control to retain the information relating to the risk area. For example, the

retention control unit **310** causes the storage unit **380** to store the information relating to the risk area. The risk area information includes a plurality of points that define the risk area. The retention control unit **310** performs a control to retain coordinate information of the plurality of points included in the risk area information. For example, the retention control unit **310** causes the storage unit **380** to store the coordinate information of the plurality of points included in the risk area information.

(42) The receiving control unit **360** performs a control to periodically receive location information of a plurality of terminals including the terminal **82**. The server **52** manages the location information of each of the plurality of terminals by storing, in the storage unit **380**, the location information of the plurality of terminals received by the control of the receiving control unit **360**.

(43) The receiving control unit **360** receives the risk area information from the vehicle **20** or the vehicle **60**. The risk area information is information that is transmitted from the vehicle **20** or the vehicle **60**, and is information for inquiring whether any of the terminals whose location information is managed by the server **52** is present in the risk area. The risk area information may include all coordinates of a plurality of vertices that define the risk area. Based on the location information of the terminal that is managed by the server **52**, the determination unit **320** determines whether any of the terminals is present in the risk area defined by a plurality of vertices included in the risk area information. When any of the terminals is determined to be present in the risk area, the transmission control unit **250** performs a control to transmit the response information to the effect that the terminal is present, to the vehicle **20** or the vehicle **60** that is a transmission source of the inquiry information.

(44) FIG. **6** shows an outline of an operation when the vehicle **20** transmits a risk area request to the server **52** according to a change in a travel path. FIG. **6** shows a process of the vehicle **20** changing a course from a road **71** to a road **72**. A server **52a** and a server **52b** correspond to the server **52** described above, and a base station **50a** and a base station **50b** correspond to the base station **50** described above. The server **52a** is a server provided mainly to provide each vehicle with the coordinate information of the risk area of the road **71**. The server **52b** is a server provided mainly to provide each vehicle with the coordinate information of the risk area of the road **72**.

(45) The transmission control unit **250** in the vehicle **20** periodically transmits the risk area request while the vehicle **20** is traveling on the road **71**. In the example of FIG. **6**, the information processing apparatus **24** of the vehicle **20** transmits the risk area request to the server **52a** during the travel on the road **71**, and receives a predetermined number of pieces of the coordinate information of the risk area on the road **71**, the coordinate information being transmitted from the server **52a**, and the information processing apparatus **24** stores the received coordinate information in the storage unit **280**. As described above, the information processing apparatus **24** uses the stored coordinate information of the risk area, and sets, in a case of reaching near the risk area, the risk area information to issue the warning or the like to the terminal **82** that is present in the risk area.

(46) The occupant of vehicle **20** operates the turn signal lamp **42** of vehicle **20**, before reaching an intersection of the road **71** and the road **72**, to move from the road **71** to the road **72**. The prediction unit **230** predicts a future travel direction of the vehicle **20**, based on the operation information of the turn signal lamp **42**, according to the operation of the turn signal lamp **42** being started. For example, when the operation of the turn signal lamp **42** for turning left is started, the prediction unit **230** predicts that the vehicle **20** will move to a left direction with respect to the current travel direction, and when the operation of the turn signal lamp **42** for turning right is started, the prediction unit **230** predicts that the vehicle **20** will move to a right direction with respect to the current travel direction. In addition, the determination unit **240** stores the current travel direction of the vehicle **20**, according to the operation of the turn signal lamp **42** being started.

(47) When the left turn of the vehicle **20** is completed and the occupant of the vehicle **20** stops the operation of the turn signal lamp **42**, the determination unit **240** starts measuring a travel distance of the vehicle **20** after the operation of the turn signal lamp **42** is stopped, according to the stop of



the operation of the turn signal lamp **42**. When the travel distance of the vehicle **20** after the operation of the turn signal lamp **42** is stopped exceeds a predetermined distance, the determination unit **240** identifies the current travel direction of the vehicle **20**, and when the current travel direction of the vehicle **20** matches the travel direction predicted by the prediction unit **230** in a predetermined range, the determination unit **240** determines that the travel path of the vehicle **20** has been changed.

(48) When the determination unit **240** determines that the travel path of the vehicle **20** has been changed, the transmission control unit **250** performs the control to transmit the risk area request to the server **52b**. When the risk area request is received from the vehicle **20**, the server **52b** transmits a predetermined number of pieces of the coordinate information of the risk area on the road **72** as the response information to the risk area request. In the vehicle **20**, when the coordinate information of the risk area on the road **72** is received by the control of the receiving control unit **260**, the received coordinate information of the risk area is stored in the storage unit **280**. Thereby, when the travel path is changed from the road **71** to the road **72**, the vehicle **20** can acquire the information of the risk area on the road **72** that is a changed destination.

(49) FIG. **7** shows an example of a flowchart relating to processing for an information processing apparatus **24** to receive coordinate information of the risk area. The processing of the present flowchart is started each time a predetermined event that triggers the transmission of the risk area request occurs.

(50) The event that triggers the transmission of the risk area request includes: an elapsed time event that occurs each time a predetermined time elapses; a “travel distance event” that occurs each time the travel distance of the vehicle **20** becomes greater than or equal to a predetermined first distance; a “location point” event that occurs by the vehicle **20** being closer to within a predetermined distance from a predetermined location point; and a “turn signal lamp ON” event that occurs by the operation of the turn signal lamp **42** being started. The elapsed time events occur, for example, every two minutes. The travel distance event occurs, for example, each time the travel distance becomes greater than or equal to a predetermined distance between 1 km and 2 km. In this way, the information processing apparatus **24** can receive the coordinate information of the risk area at least each time a predetermined time elapses. In addition, the information processing apparatus **24** can receive the coordinate information of the risk area at least each time the travel distance of the vehicle **20** becomes greater than or equal to the first distance. In addition, the information processing apparatus **24** can receive the coordinate information of the risk area each time the vehicle **20** becomes closer to within a predetermined distance from a predetermined inquiry location point.

(51) If the event that triggers the transmission of the risk area request occurs, the determination unit **240** determines a type of event that has occurred, in **S702**. If the elapsed time event or the travel distance event occurs, the communication apparatus **48** transmits, by the control of the transmission control unit **250**, the risk area request to the server **52**, in **S704**. In **S706**, the communication apparatus **48** receives, by the control of the receiving control unit **260**, the coordinate information of the risk area which is transmitted from the server **52**. In **S708**, the storage unit **380** stores the coordinate information of the risk area. This makes it possible to receive the latest coordinate information of the risk area, periodically or each time the travel distance of the vehicle **20** becomes greater than or equal to the first distance.

(52) If the turn signal lamp ON event is determined to have occurred in the determination of **S702**, the determination unit **240** stores the current travel direction of the vehicle **20**, in **S712**.

Subsequently, in **S712**, the prediction unit **230** predicts the future travel direction of the vehicle **20** based on the operation information of the turn signal lamp **42**. For example, the prediction unit **230** predicts the future travel direction of the vehicle **20** from the current travel direction stored in **S710**, and a relative change direction for the future travel direction which is determined from the operation information of the turn signal lamp **42**.

(53) Subsequently, in S714, the determination unit 240 waits until the operation of the turn signal lamp 42 is stopped. When the turn signal lamp 42 is stopped, the determination unit 240 waits until the travel distance of the vehicle 20 after the operation of the turn signal lamp 42 is stopped becomes greater than or equal to a predetermined second distance, in S716. For example, the determination unit 240 waits until the travel distance of the vehicle 20 after the operation of the turn signal lamp 42 is stopped becomes 10 m or more. It should be noted that the second distance is shorter than the first distance.

(54) If the travel distance of the vehicle 20 after the operation of the turn signal lamp 42 is stopped is determined to become greater than or equal to the second distance by S716, the determination unit 240 determines whether the travel direction of the vehicle 20 has been changed to the predicted direction, in S718. For example, if the current travel direction of the vehicle 20 matches the travel direction predicted in S712, the determination unit 240 determines that the travel direction of the vehicle 20 has been changed to the predicted direction. As an example, if a difference between the current travel direction of the vehicle 20 and the travel direction predicted in S712 is in a predetermined range, the current travel direction of the vehicle 20 may be determined to match the travel direction predicted in S712.

(55) If the travel direction of the vehicle 20 is determined to have been changed as predicted in S718, the coordinate information of the risk area received in the past and stored in the storage unit 380 is discarded, in S720, and processing proceeds to S704. This makes it possible for the information processing apparatus 24 to transmit the risk area request, to receive the coordinate information of the risk area, and to store the coordinate information of the risk area, regardless of whether the elapsed time event or the travel distance event occurs, at a timing different from a timing at which the elapsed time event or the travel distance event occurs. On the other hand, if the travel direction of the vehicle 20 is determined not to be changed as predicted in S718, the processing of the present flowchart is ended without transmitting the risk area request.

(56) In this way, with the information processing apparatus 24, it is possible to transmit the risk area request when the travel path of the vehicle 20 is changed. On the other hand, it is possible not to transmit the risk area request when the vehicle 20 travels along a road and makes a turn, or when a lane is changed, or the like, and thus it is possible to suppress a frequent transmission of the risk area request. It should be noted that the above embodiment describes the operation of the information processing apparatus 24 transmitting the risk area request by using a scene in which the vehicle 20 turns at an intersection; however, the embodiment can be applied not only to the case of turning at an intersection, but also to a case of making a U-turn on the same road to advance in an opposite direction.

(57) As described above, with the information processing apparatus 24, it is possible to receive, from the server 52, the coordinate information of the risk area of a path to a changed destination, according to the vehicle 20 changing the travel path. Therefore, it is possible to efficiently acquire the coordinate information of the risk area without shortening a time interval for periodically acquiring the coordinate information of the risk area. This makes it possible to reduce a load on the process of the server 52 to provide the coordinate information of the risk area to the vehicle 20.

(58) It should be noted that the communication between the information processing apparatus 24 and the server 52 may be performed by a communication system conforming to Cellular-V2X. The Cellular-V2X includes communication systems such as LTE-V2X PC5 and 5G-V2X PC5. In another embodiment, the communication between the information processing apparatus 24 and the server 52 may adopt a mode using Wi-Fi (registered trademark) or DSRC (Dedicated Short Range Communications). For the communication between the information processing apparatus 24 and the server 52, any communication system such as Bluetooth (registered trademark) may be adopted in addition to the Cellular-V2X, the DSRC (registered trademark), or the like. The information processing apparatus 24 may communicate with the server 52 by using a communication infrastructure included in ITS (Intelligent Transport Systems: an advanced road transportation

system).

(59) It should be noted that the vehicle **20** and the vehicle **60** may be examples of transportation equipment. Examples of the transportation equipment may include an automobile such as a passenger car and a bus, a saddle riding type vehicle, and a bicycle, or the like. In addition, in the present embodiment described above, the system **10** functions as the system for issuing the warning when the terminal **82** is present in the risk area, but may function as a system for issuing the warning when any communication apparatus whose location information is able to be managed by the server **52**, other than the terminal **82**, is present. Such a communication apparatus may be included in any moving object such as the automobile, the saddle riding type vehicle, and the bicycle.

(60) As described above, the present embodiment has been made by taking into account that the load on the process of the server **52** to provide the coordinate information of the risk area to the vehicle **20**, increases, and a purpose of the present embodiment is to enhance traffic safety and to suppress deterioration of traffic smoothness.

(61) FIG. **8** shows an example of a computer **2000** in which a plurality of embodiments of the present invention may be entirely or partially embodied. A program that is installed in the computer **2000** can cause the computer **2000** to function as an apparatus such as the information processing apparatus **24** according to the embodiment or each unit of the apparatus, or can cause the computer **2000** to function as a server such as the server **52** according to the embodiment or each unit of the server, or cause the computer **2000** to execute an operation that is associated with the apparatus, each unit of the apparatus, the server, or each unit of the server, and/or can cause the computer **2000** to execute processes or steps of the processes according to the embodiment. Such a program may be executed by a CPU **2012** in order to cause the computer **2000** to execute a specific operation associated with some or all of the blocks in the processing procedure and the block diagram described in the present specification.

(62) The computer **2000** according to the present embodiment includes the CPU **2012** and a RAM **2014**, which are connected mutually by a host controller **2010**. The computer **2000** also includes a ROM **2026**, a flash memory **2024**, a communication interface **2022**, and an input/output chip **2040**. The ROM **2026**, the flash memory **2024**, the communication interface **2022**, and the input/output chip **2040** are connected to the host controller **2010** via an input/output controller **2020**.

(63) The CPU **2012** operates according to the program stored in the ROM **2026** and the RAM **2014**, thereby controlling each unit.

(64) The communication interface **2022** communicates with other electronic devices via a network. The flash memory **2024** stores the program and data used by the CPU **2012** in the computer **2000**. The ROM **2026** stores a boot program or the like executed by the computer **2000** during activation, and/or a program depending on hardware of the computer **2000**. The input/output chip **2040** may also connect, to the input/output controller **2020**, a variety of input/output units such as a keyboard, a mouse, and a monitor, via input/output ports such as a serial port, a parallel port, a keyboard port, a mouse port, a monitor port, a USB port, and an HDMI (registered trademark) port.

(65) A program is provided via a computer-readable medium such as a CD-ROM, a DVD-ROM, or a memory card, or a network. The RAM **2014**, the ROM **2026**, or the flash memory **2024** is an example of the computer-readable medium. The program is installed in the flash memory **2024**, the RAM **2014**, or the ROM **2026**, and is executed by the CPU **2012**. Information processing written in the program is read by the computer **2000**, thereby resulting in cooperation between the program, and the above-described various types of hardware resources. An apparatus or method may be constituted by implementing the operations or processing on information according to the use of the computer **2000**.

(66) For example, when communications are performed between the computer **2000** and external devices, the CPU **2012** may execute a communication program loaded in the RAM **2014**, and instruct the communication interface **2022** to execute communication processing on based on the

processes written in the communication program. The communication interface **2022**, under the control of the CPU **2012**, reads out transmission data stored in a transmission buffer processing region provided in a recording medium such as the RAM **2014** and the flash memory **2024**, transmits the read transmission data to the network, and writes reception data received from the network into a reception buffer processing region or the like provided on the recording medium. (67) Also, the CPU **2012** may cause all or required portions of a file or a database stored in the recording medium such as the flash memory **2024** to be read by the RAM **2014**, and perform various types of processes on the data on the RAM **2014**. The CPU **2012**, then, writes back the processed data into the recording medium.

(68) Various types of programs and various types of information such as data, tables, and databases may be stored in the recording medium to undergo information processing. The CPU **2012** may perform, on the data read from the RAM **2014**, various types of processing including various types of operations, information processing, conditional judgments, conditional branching, unconditional branching, information retrievals/replacements, and the like described in the present specification and specified by an instruction sequence of the program, to write back the result into the RAM **2014**. Also, the CPU **2012** may search for information in a file, a database, etc., in the recording medium. For example, when a plurality of entries each having an attribute value of a first attribute respectively associated with an attribute value of a second attribute is stored in the recording medium, the CPU **2012** may search for, from among the plurality of entries, an entry in which the attribute value of the first attribute is specified and that match with a condition, read the attribute value of the second attribute stored in the entry, and thereby obtain the attribute value of the second attribute associated with the first attribute meeting a predetermined condition.

(69) The program or software module described above may be stored in a computer-readable medium on the computer **2000** or near the computer **2000**. A recording medium such as a hard disk or a RAM provided in a server system connected to a dedicated communication network or the Internet can be used as a computer-readable medium. The program stored in the computer-readable medium may be provided to the computer **2000** via the network.

(70) When the computer **2000** functions as the control unit **200**, a program that is installed in the computer **2000** and causes the computer **2000** to function as the control unit **200** may work on the CPU **2012** and the like to cause the computer **2000** to function as each unit of the control unit **200**. Information processing written in these programs functions as each unit of the control unit **200** that is specific means by which software and the above-described various hardware resources cooperate by being read by the computer **2000**. Then, by the specific means realizing calculation or processing of information according to a purpose of a use of the computer **2000** in the present embodiment, the unique control unit **200** according to the purpose of the use is constructed.

(71) When the computer **2000** functions as the control unit **300**, a program that is installed in the computer **2000** and causes the computer **2000** to function as the control unit **300** may work on the CPU **2012** and the like to cause the computer **2000** to function as each unit of the control unit **300**. Information processing written in these programs functions as each unit of the control unit **300** that is specific means by which software and the above-described various hardware resources cooperate by being read by the computer **2000**. Then, by the specific means realizing calculation or processing of information according to the purpose of the use of the computer **2000** in the present embodiment, the unique control unit **300** according to the purpose of the use is constructed.

(72) A variety of embodiments have been described with reference to the block diagram and the like. In the block diagram, each block may represent (1) a step of a process in which an operation is performed, or (2) each unit of the apparatus having a role of performing the operation. Certain steps and “units” may be implemented by dedicated circuitry, programmable circuitry supplied with computer-readable instructions stored on computer-readable media, and/or processors supplied with computer-readable instructions stored on computer-readable media. The dedicated circuitry may include a digital and/or analog hardware circuit, or may include an integrated circuit (IC),

and/or a discrete circuit. The programmable circuitry may include a reconfigurable hardware circuit including: logical AND, logical OR, exclusive OR (XOR), negative AND (NAND), negative OR (NOR), and other logical operation; and a memory element such as a flip-flop, a register, a field-programmable gate array (FPGA), a programmable logic array (PLA); and so on.

(73) A computer-readable medium may include any tangible device that can store instructions to be executed by a suitable device, and as a result, the computer-readable medium having the instructions stored thereon constitutes at least a part of an article of manufacture including instructions which can be executed to create means for performing operations specified in the processing procedures or block diagrams. Examples of the computer-readable medium may include an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, and the like. Specific examples of the computer-readable medium may include a floppy (registered trademark) disk, a diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an electrically erasable programmable read-only memory (EEPROM), a static random access memory (SRAM), a compact disc read-only memory (CD-ROM), a digital versatile disc (DVD), a Blu-ray (registered trademark) disc, a memory stick, an integrated circuit card, or the like.

(74) The computer-readable instruction may include either of source code or object code written in any combination of one or more programming languages including: an assembler instruction, an instruction-set-architecture (ISA) instruction, a machine instruction, a machine dependent instruction, a microcode, a firmware instruction, state-setting data; or an object oriented programming language such as Smalltalk (registered trademark), JAVA (registered trademark), C++, or the like; and a conventional procedural programming language such as a “C” programming language or a similar programming language.

(75) Computer-readable instructions may be provided to a general-purpose computer, a special-purpose computer, or a processor or programmable circuitry of other programmable data processing apparatus, locally or via a local area network (LAN), a wide area network (WAN) such as the internet, and the computer-readable instructions may be executed in order to result in means for executing operations specified by the described processing procedure or the block diagram. Examples of the processor include a computer processor, a processing unit, a microprocessor, a digital signal processor, a controller, a microcontroller, and the like.

(76) While the embodiments of the present invention have been described, the technical scope of the present invention is not limited to the above-described embodiments. It is apparent to persons skilled in the art that various alterations or improvements can be made to the above-described embodiments. It is also apparent from the description of the claims that the embodiments to which such alterations or improvements are made can be included in the technical scope of the present invention.

(77) The operations, procedures, steps, and stages of each process performed by an apparatus, system, program, and method shown in the claims, embodiments, or diagrams can be performed in any order as long as the order is not indicated by “prior to,” “before,” or the like and as long as the output from a previous process is not used in a later process. Even if the process flow is described using phrases such as “first” or “next” in the claims, embodiments, or diagrams, it does not necessarily mean that the process must be performed in this order.

#### EXPLANATION OF REFERENCES

(78) **10** system **20, 60, 90** vehicle **21** radar **22** camera **24** information processing apparatus **25** GNSS receiving unit **26** vehicle speed sensor **28** yaw rate sensor **29** sensor **40** information output apparatus **42** turn signal lamp **48** communication apparatus **50** base station **52** server **64** information processing apparatus **70, 71, 72** road **80** person **82** terminal **110, 120** area **111, 112, 113, 114, 123, 124** vertex **200** control unit **208** control unit **210** coordinate information acquisition unit **220** risk area identification unit **230** prediction unit **240** determination unit **250** transmission

control unit **260** receiving control unit **280** storage unit **300** control unit **310** retention control unit **320** determination unit **348** communication apparatus **350** transmission control unit **360** receiving control unit **380** storage unit **2000** computer **2010** host controller **2012** CPU **2014** RAM **2020** input/output controller **2022** communication interface **2024** flash memory **2026** ROM **2040**: input/output chip

## Claims

1. An information processing apparatus mounted on a vehicle, comprising: one or more processors, wherein the one or more processors are configured to: predict a change in a movement direction of the vehicle; determine whether the movement direction of the vehicle has been changed to the direction predicted; perform a control to receive information indicating a risk area ahead in the movement direction of the vehicle, from a server configured to retain information relating to a risk area, when the movement direction of the vehicle is determined to have been changed to the predicted direction; predict the change in the movement direction of the vehicle based on operation information of a turn signal lamp included in vehicle; and determine that the movement direction of the vehicle has been changed to the predicted direction, in a case where an operation of the turn signal lamp is stopped and then the vehicle travels a distance greater than or equal to a predetermined distance.
2. The information processing apparatus according to claim 1, wherein the one or more processors are further configured to perform a control to receive the information indicating the risk area at a predetermined time interval, and perform a control to receive the information indicating the risk area, at a timing other than a timing at which the information indicating the risk area is received at the predetermined time interval, when the movement direction of the vehicle is determined to have been changed to the predicted direction.
3. The information processing apparatus according to claim 1, wherein the one or more processors are further configured to perform a control to discard already received information indicating the risk area, and receive the information indicating the risk area, when the movement direction of the vehicle is determined to have been changed to the predicted direction.
4. The information processing apparatus according to claim 1, wherein the one or more processors are further configured to store the movement direction of the vehicle in a case where the operation of the turn signal lamp is started, and determine that the movement direction of the vehicle has been changed to the predicted direction, in a case where the operation of the turn signal lamp is stopped and then the vehicle travels a predetermined distance or more in a direction different from the stored movement direction.
5. The information processing apparatus according to claim 1, wherein the one or more processors are further configured to identify a risk area outside the vehicle; and perform a control to transmit risk area information indicating the identified risk area, to the server.
6. The information processing apparatus according to claim 1, wherein the server is a mobile edge computing (MEC) server.
7. A vehicle comprising the information processing apparatus according to claim 1.
8. A system comprising: the vehicle according to claim 7; and the server.
9. An information processing apparatus mounted on a vehicle comprising: one or more processors, wherein the one or more processors are configured to: predict a change in a movement direction of the vehicle; determine whether the movement direction of the vehicle has been changed to the direction predicted; perform a control to receive information indicating a risk area ahead in the movement direction of the vehicle, from a server configured to retain information relating to a risk area, when the movement direction of the vehicle is determined to have been changed to the predicted direction; predict the change in the movement direction of the vehicle based on operation information of a turn signal lamp included in the vehicle; and determine that the movement

direction of the vehicle has been changed to the predicted direction, in a case where an operation of the turn signal lamp is stopped and then the vehicle travels a predetermined time or more.

10. The information processing apparatus according to claim 9, wherein the one or more processors are further configured to store the movement direction of the vehicle in a case where the operation of the turn signal lamp is started, and determine that the movement direction of the vehicle has been changed to the predicted direction, in a case where the operation of the turn signal lamp is stopped and then the vehicle travels a predetermined time or more in a direction different from the stored movement direction.

11. The information processing apparatus according to claim 9, wherein the one or more processors are further configured to perform a control to receive the information indicating the risk area at a predetermined time interval, and perform a control to receive the information indicating the risk area, at a timing other than a timing at which the information indicating the risk area is received at the predetermined time interval, when the movement direction of the vehicle is determined to have been changed to the predicted direction.

12. The information processing apparatus according to claim 9, wherein the one or more processors are further configured to perform a control to discard already received information indicating the risk area, and receive the information indicating the risk area, when the movement direction of the vehicle is determined to have been changed to the predicted direction.

13. The information processing apparatus according to claim 9, wherein the one or more processors are further configured to identify a risk area outside the vehicle; and perform a control to transmit risk area information indicating the identified risk area, to the server.

14. The information processing apparatus according to claim 9, wherein the server is a mobile edge computing (MEC) server.

15. A vehicle comprising the information processing apparatus according to claim 9.

16. A system comprising: the vehicle according to claim 15; and the server.

17. An information processing method performed by an information processing apparatus mounted on a vehicle, comprising: predicting a change in a movement direction of the vehicle; determining whether the movement direction of the vehicle has been changed to the predicted direction; and performing a control to receive information indicating a risk area ahead in the movement direction of the vehicle, from a server configured to retain information relating to a risk area, when the movement direction of the vehicle is determined to have been changed to the predicted direction; predicting the change in the movement direction of the vehicle based on operation information of a turn signal lamp included in the vehicle; and determining that the movement direction of the vehicle has been changed to the predicted direction, in a case where an operation of the turn signal lamp is stopped and then the vehicle travels a distance greater than or equal to a predetermined distance.

18. A non-transitory computer-readable storage medium having stored thereon a program that causes a computer mounted on a vehicle to function as: a receiving control unit configured to perform a control to receive information indicating a risk area ahead in a movement direction of the vehicle, from a server configured to retain information relating to a risk area; a prediction unit configured to predict a change in the movement direction of the vehicle; and a determination unit configured to determine whether the movement direction of the vehicle has been changed to the direction predicted by the prediction unit, wherein the receiving control unit is configured to perform a control to receive the information indicating the risk area, when the determination unit determines that the movement direction of the vehicle has been changed to the direction predicted by the prediction unit; the prediction unit is configured to predict the change in the movement direction of the vehicle based on operation information of a turn signal lamp included in the vehicle; and the determination unit is configured to determine that the movement direction of the vehicle has been changed to the direction predicted by the prediction unit, in a case where an operation of the turn signal lamp is stopped and then the vehicle travels a distance greater than or

equal to a predetermined distance.

19. An information processing method performed by an information processing apparatus mounted on a vehicle, comprising: predicting a change in a movement direction of the vehicle; determining whether the movement direction of the vehicle has been changed to the predicted direction; performing a control to receive information indicating a risk area ahead in the movement direction of the vehicle, from a server configured to retain information relating to a risk area, when the movement direction of the vehicle is determined to have been changed to the predicted direction; predicting the change in the movement direction of the vehicle based on operation information of a turn signal lamp included in the vehicle; and determining that the movement direction of the vehicle has been changed to the predicted direction, in a case where an operation of the turn signal lamp is stopped and then the vehicle travels a predetermined time or more.

20. A non-transitory computer-readable storage medium having stored thereon a program that causes a computer mounted on a vehicle to function as: a receiving control unit configured to perform a control to receive information indicating a risk area ahead in a movement direction of a vehicle, from a server configured to retain information relating to a risk area; a prediction unit configured to predict a change in the movement direction of the vehicle; and a determination unit configured to determine whether the movement direction of the vehicle has been changed to the direction predicted by the prediction unit, wherein the receiving control unit is configured to perform a control to receive the information indicating the risk area, when the determination unit determines that the movement direction of the vehicle has been changed to the direction predicted by the prediction unit; the prediction unit is configured to predict the change in the movement direction of the vehicle based on operation information of a turn signal lamp included in the vehicle; and the determination unit is configured to determine that the movement direction of the vehicle has been changed to the direction predicted by the prediction unit, in a case where an operation of the turn signal lamp is stopped and then the vehicle travels a predetermined time or more.

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